Substrate Rigidity and the K-Theoretic Obstruction to α-Derivation in Principia Fractalis: a Machine-Checked Joint Statement in Lean 4

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September 10, 2026

Abstract

We report a machine-checked *joint statement* about the algebraic substrate T_∞ of the *Principia Fractalis* research program [6]. The joint statement pairs a positive uniqueness result (new to this paper) with a negative derivability result (previously kernel-verified in the parent project), and it is offered as the honest scientific summary of what the program has established about its substrate axis through 2026-09-10.

(1) *Substrate rigidity, new.* We give the first machine-checked formalization of the *uniqueness half* of Glimm's 1960 classification of uniformly hyperfinite (UHF) C^*-algebras, specialized to supernatural number 3^∞, in the Lean 4 proof assistant (uniqueness for the exact 3^∞ tower; refinement-invariance and the distinct-supernatural-number direction are not formalized). The main theorem `T_infinity_rigidity` asserts that any C^*-algebra carrying a *Substrate3Inf* witness — a directed tower of $M_{3^k}(\mathbb{C})$ subalgebras with dense union — is $*$-isomorphic to the distinguished object T_∞, constructed as the norm completion of the algebraic direct limit of the ternary tower. Uniqueness of the (continuous, unital) tracial linear functional is *derived* as a consequence of this rigidity (`Substrate3Inf.trace_unique`), not assumed as a hypothesis — so the dense ternary tower alone forces the substrate. All seven load-bearing declarations (including the derived `Substrate3Inf.trace_unique`) audit to exactly {`propext`, `Classical.choice`, `Quot.sound`}, with no `sorry`, no project axioms, and no `native_decide`.

(2) *The α-derivation obstruction, cited.* We pair this with a companion theorem, proved in the parent Principia Fractalis codebase and cited here as `pi_not_ktheoretic_ratio` (Cohen, r332, 2026-09-07), asserting that for all $a, b \in \mathbb{Z}[1/3]$, $b/a \neq \pi$. Combined with the earlier r123 result that the substrate's K-theoretic trace range is exactly $\mathbb{Z}[1/3]$, this establishes that seven of the nine α-values enumerated by the program (all irrationals, plus the ratio $\alpha_{\mathrm{NS}}/\alpha_{\mathrm{RH}} = \pi$) cannot be derived from the substrate through its K-theoretic channel — the only channel the substrate mathematically possesses.

The paired reading is that Principia Fractalis's substrate is now a uniquely-determined mathematical object, and that its K-theoretic invariant provably underdetermines the framework's α-skeleton. Deriving the α-values — or explicitly declining to — must therefore happen at a higher layer than the substrate. We discuss the implications in Section 8. The formalization contributes several primitives absent from `mathlib4`, listed in Section 6 and offered as candidate upstream contributions.

1 Introduction

The theory of *uniformly hyperfinite* (UHF) C^*-algebras was inaugurated by Glimm [1] in 1960. A UHF algebra is the norm completion of an inductive limit of finite matrix algebras under

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unital $*$-embeddings; Glimm proved that two UHF algebras are $*$-isomorphic if and only if their *supernatural numbers* — formal products $\prod_p p^{n_p}$ with $n_p \in \mathbb{N} \cup \{\infty\}$, tracking the multiplicities of the matrix sizes — coincide. The proof strategy is the archetypal "Elliott intertwining" [2], later generalized by Bratteli, Elliott, Rørdam, and others to a substantial classification programme.

Despite the classical status of Glimm's result, it has not, until now, been formalized in any interactive theorem prover we are aware of. We verified this against `mathlib4` at commit HEAD (September 9, 2026; this novelty check is to be re-run at submission, as mathlib's star-algebra apparatus — e.g. `conjStarAlgAut` — continues to evolve), which contains substantial C^*-algebra machinery (`CStarAlgebra`, `StarAlgHom`, `StarAlgEquiv`, orthonormal-basis and reindex apparatus) but no `UHF` class, no `Bratteli` diagrams, and no direct-limit theory adapted to non-commutative C^*-algebras. We also confirmed the absence in `Isabelle/HOL` (Archive of Formal Proofs), `Coq/Rocq` (MathComp, coq-community), `Agda` (agda-stdlib), and `Lean 3`.

The present work formalizes Glimm's classification *specialized to supernatural number* 3^∞ — that is, UHF algebras of Bratteli type $M_3 \otimes M_3 \otimes M_3 \otimes \cdots$. This specialization is motivated by an ongoing research program (Section 8) in which a distinguished object of that isomorphism class, denoted here T_∞, plays the role of a substrate. Rather than parameterize over all supernatural numbers — which would require a full formalization of the supernatural-number arithmetic and its associated Bratteli-diagram theory — we build the specialization directly, with the ternary structure hard-wired into the definitions. Two components of Glimm's full classification are deliberately out of scope and are *not* claimed here: *refinement-invariance* (that a tower whose k-th level is $M_{9^k}(\mathbb{C})$, or any other presentation with the same supernatural number 3^∞, yields the same algebra), and the *completeness* direction (that *distinct* supernatural numbers yield non-isomorphic algebras). What is formalized is the uniqueness statement for the fixed 3^∞ tower: any two C^*-algebras carrying the `Substrate3Inf` structure are $*$-isomorphic.

Contributions

1. A Lean 4 `structure Substrate3Inf` characterizing UHF algebras of type 3^∞ by four fields — a tower of subalgebras, each $*$-isomorphic to $M_{3^k}(\mathbb{C})$, with monotone inclusions and dense union (Definition 3.1); unique traciality is *derived*, not assumed (Theorem 3.4).

2. The main classification theorem `T_infinity_rigidity` (Theorem 3.5), with a proof organized into four arcs (Section 5): block-diagonal embedding (C1), Noether–Skolem uniqueness (C2), completion universal property (C3), and a level-preserving recursion (C4; a one-sided, degenerate case of Elliott intertwining — both towers share the dimension sequence 3^k, so no genuine back-and-forth is required).

3. An explicit statement of the paired *K-theoretic obstruction* to α-derivation (Theorem 4.2, Section 4), extracted from the parent Principia Fractalis codebase and set alongside the uniqueness result to give the honest joint statement about the substrate axis.

4. Six auxiliary primitives absent from mathlib at time of writing, listed in Section 6 and offered as candidate upstream contributions: `conjByUnitary`, `reindexStarAlgEquiv`, a star-preserving completion lift, a non-commutative C^*-direct-limit apparatus, and the `IsTracialLinearFunctio` predicate with its uniqueness lemma.

5. A machine-checked axiom-audit chain: all seven load-bearing declarations of the uniqueness result — including the derived trace-uniqueness theorem — together with the two cited declarations of the obstruction result, reduce to {`propext`, `Classical.choice`, `Quot.sound`} (Section 7).

The complete formalization is a single Lean file, `PF/SubstrateRigidity.lean`, comprising approximately 2,500 lines. It resides in the public `Principia-Fractalis` repository on GitHub (branch `master`, commit `398e3dfd`); see Section 11 for access.

On authorship and collaboration

The formalization in this paper was carried out with the assistance of the Anthropic Claude Opus language model, following the collaboration pattern established by Anthropic's formalization of Fermat's Last Theorem in September 2026 [9]. The sole author of this work is Pablo Cohen; the language model is a tool, not an author, and is credited here and in the acknowledgments rather than on the author line. All *mathematical* content was authored, reviewed, and gated by the author; the Lean 4 *proof engineering* was farmed to a coordinated series of prover-agent sessions under an explicit statement-card discipline. The kernel is the arbiter; no proof was accepted without an axiom-audit line reading exactly {`propext`, `Classical.choice`, `Quot.sound`}.

2 Background: UHF algebras and Glimm's classification

Throughout, $\mathbb{C}$ denotes the complex field and $M_n(\mathbb{C})$ the C^*-algebra of $n \times n$ complex matrices under the operator norm inherited from the standard inner-product structure on $\mathbb{C}^n$.

Definition 2.1 (UHF algebra)**.** A *uniformly hyperfinite* C^*-algebra A is a unital C^*-algebra given by the norm-closure of an inductive limit

$$A_0 \hookrightarrow A_1 \hookrightarrow A_2 \hookrightarrow \cdots$$

where each A_k is $*$-isomorphic to a full matrix algebra $M_{n_k}(\mathbb{C})$ over $\mathbb{C}$, and each inclusion is a unital $*$-embedding.

Definition 2.2 (Supernatural number of a UHF algebra)**.** Given a UHF algebra A presented as above, the *supernatural number* of A is the formal product $\prod_p p^{n_p}$, where $n_p \in \mathbb{N} \cup \{\infty\}$ is the supremum of the p-adic valuations of the dimensions n_k.

Theorem 2.3 (Glimm 1960)**.** *Two UHF algebras are $*$-isomorphic if and only if their supernatural numbers coincide.*

Glimm's proof of the "$\Leftarrow$" direction proceeds by an Elliott back-and-forth intertwining argument: given two UHF towers $(A_k)_{k\in\mathbb{N}}$ and $(B_k)_{k\in\mathbb{N}}$ with matching supernatural data, one constructs an increasing family of unital $*$-embeddings $A_k \hookrightarrow B_{k+n_k}$ (and vice versa) using a lemma of Noether–Skolem type [3, 4]: any two unital $*$-homomorphisms $M_n(\mathbb{C}) \to M_{kn}(\mathbb{C})$ are unitarily conjugate. Passage to the norm limit then yields the desired $*$-isomorphism.

The present work formalizes this argument in the specialization $\prod_p p^{n_p} = 3^\infty$, that is, $A_k \cong M_{3^k}(\mathbb{C})$.

3 The Substrate3Inf structure and the theorem

3.1 Characterization

We formalize the class of UHF algebras of type 3^∞ as a structure over an arbitrary C^*-algebra `A`. The four fields collectively pin down the isomorphism class. Unique traciality is *not* among them: it is derived (Theorem 3.4), not assumed.

Definition 3.1 (Substrate3Inf).

```
structure Substrate3Inf (A : Type*) [CStarAlgebra A] where
  tower : ℕ→StarSubalgebra ℂA
  tower_matrix : ∀k, Nonempty
                  (tower k ≃⋆ₐ[ℂ] Matrix (Fin (3^k)) (Fin (3^k)) ℂ)
  tower_mono : ∀k, tower k ≤tower (k+1)
  tower_dense : Dense (((⨆ k, tower k : StarSubalgebra ℂA) : Set A))
```

The fields are:

`tower`: a sequence of $\mathbb{C}$-star-subalgebras $A_k \subseteq A$;

`tower_matrix`: each A_k is $*$-isomorphic to $M_{3^k}(\mathbb{C})$; the assertion is stated as *some such iso exists* (`Nonempty`), not as a distinguished datum, so as not to impose a coherence hypothesis that Glimm's classification is meant to derive;

`tower_mono`: the tower is monotone under inclusion;

`tower_dense`: the norm-closure of $\bigcup_k A_k$ is all of A; this is the "completion" condition that distinguishes a UHF algebra from the bare algebraic direct limit.

The predicate `IsTracialLinearFunctional` is defined *locally* to the file (mathlib does not currently offer a named tracial-state class) as follows:

Definition 3.2 (Tracial linear functional).

```
structure IsTracialLinearFunctional
    (A : Type*) [CStarAlgebra A] (φ : A →ℂ) : Prop where
  continuous : Continuous φ
  add : ∀x y : A, φ(x + y) = φx + φy
  smul : ∀(c : ℂ) (x : A), φ(c •x) = c * φx
  tracial : ∀x y : A, φ(x * y) = φ(y * x)
  unital : φ1 = 1
```

Remark 3.3 (Semantic caveat)**.** The predicate `IsTracialLinearFunctional` is *strictly weaker* than the standard notion of a *tracial state*: it omits positivity ($\phi(x^*x) \geq 0$) and star-hermitianness ($\phi(x^*) = \overline{\phi(x)}$). Both of these hold automatically on the completion T_∞ (via the projection-trace calculation $\tau(p) = \mathrm{card}(p)/3^k$ for projections at level k), but we do not require them at the level of the structure. This choice matches the existing kernel-verified predicate `IsTracialState` in the Principia Fractalis codebase and is sufficient for the Elliott specialization to 3^∞, in which the projection-trace pairing suffices to pin the classifying invariant. An upgrade path adding these axioms is documented in the source; we do not depend on it here.

Theorem 3.4 (Trace uniqueness, derived; `Substrate3Inf.trace_unique`)**.** *Every* ***`Substrate3Inf`*** *witness carries a* unique *continuous unital tracial linear functional:*

```
theorem Substrate3Inf.trace_unique (h : Substrate3Inf A) :
    ∃! τ: A →ℂ, IsTracialLinearFunctional A τ
```

Unique traciality is therefore a *consequence* of the four structural fields, not a hypothesis. In an earlier version of the development (commit `18f55a14`) it was carried as a fifth field; an audit (r337, 2026-09-17) found that no proof consumed it — the recursion of arc C4 reads only `tower_matrix` and `tower_dense` — so it was removed and is recovered here by transport along the rigidity isomorphism. This strictly *strengthens* `T_infinity_rigidity`: a dense ternary matrix tower *alone* forces the substrate, and the unique trace comes out the other end, removing the objection that the hypothesis would otherwise encode half of the conclusion. The proof is kernel-clean (Section 7).

3.2 The concrete substrate

The distinguished representative of the isomorphism class is constructed in Pablo Cohen's ongoing formalization project, and is provided to us as an existing kernel-verified object. Concretely, `TimelessFieldCompletion` is defined as the uniform-space completion of an algebraic ternary direct limit:

```
abbrev TimelessFieldCompletion : Type :=
  UniformSpace.Completion TimelessFieldRing
```

where `TimelessFieldRing` is the direct limit of $M_{3^k}(\mathbb{C})$ under the block-diagonal connecting maps $x \mapsto x \otimes I_3$. The construction, together with its equipment as a C^*-algebra, is the content of a series of prior lemmas in the parent project's source tree (files `SubstrateDirectLimit.lean`, `SubstrateBase3RingHom.lean`, `SubstrateTimelessFieldNorm.lean`, `SubstrateTimelessFieldCompletion.lean`, `SubstrateTraceUniqueness.lean`), all verified against the mathlib three kernel axioms. That `TimelessFieldCompletion` is itself an inhabitant of `Substrate3Inf` is established in this work as `substrate3Inf_TimelessFieldCompletion`.

3.3 Main theorem

Theorem 3.5 (Substrate rigidity for 3^∞)**.**

```
theorem T_infinity_rigidity
    (A : Type*) [CStarAlgebra A]
    (h : Substrate3Inf A) :
    Nonempty (A ≃⋆ₐ[ℂ] TimelessFieldCompletion)
```

Any C^*-algebra carrying a `Substrate3Inf` witness is $*$-isomorphic to `TimelessFieldCompletion`.

The proof is by a one-sided, level-preserving recursion — a degenerate case of Elliott intertwining in which both towers carry the identical dimension sequence 3^k, so no genuine back-and-forth (zig-zag between the two towers) occurs — decomposed into four arcs in the source (C1–C4); see Section 5.

Corollary 3.6 (Uniqueness up to isomorphism)**.** *The isomorphism class of C^*-algebras satisfying `Substrate3Inf` contains exactly one element up to $*$-isomorphism, namely `TimelessFieldCompletion`.*

Proof. Immediate from Theorem 3.5 applied twice. □

4 The K-theoretic obstruction to α-derivation

The substrate T_∞ is a C^*-algebra with a distinguished tracial linear functional τ (continuous and unital; on T_∞ it is in fact a state, though the uniqueness in Theorem 3.4 is proved among the larger class of continuous unital tracial linear functionals, see Remark 3.3). Its *classifying invariant*, in the sense of the Elliott programme, is the pointed pre-ordered K_0 group $(K_0(T_\infty), K_0^+, [1]_0)$ together with the trace pairing. For the UHF algebra of type 3^∞, this invariant simplifies dramatically: classically, the pairing $\tau_* : K_0(T_\infty) \to \mathbb{R}$ takes values in $\mathbb{Z}[1/3]$, the dense subring of ternary-denominator rationals [1, 2]. We emphasize that K_0 itself is *not* constructed in the present Lean development; this K_0 identity is cited as classical mathematics, and what is kernel-verified here is the finite-level projection-trace computation that realizes its range (Theorem 4.1).

This is not new mathematics; it is a corollary of the classical Elliott–Glimm classification. What *is* new (or at least, what has not previously been packaged in the present form) is a machine-checked pairing of this invariant range with an explicit non-derivability result. Both directions — the invariant range and the non-derivability — are kernel-verified in the parent Principia Fractalis codebase and were established before the present formalization work; we quote them here as inputs.

Theorem 4.1 (Finite-level projection traces; Cohen, r123, 2026)**.** *For any projection p at level k (a diagonal characteristic-vector matrix in $M_{3^k}(\mathbb{C})$), the trace takes the value $\tau(p) = |S|/3^k \in \mathbb{Z}[1/3] \cap [0,1]$, where $S \subseteq \{1,\ldots,3^k\}$ is the corresponding support. Hence every trace value realized on a finite-level projection lies in $\mathbb{Z}[1/3]$.*

The Lean source is at `PF/AlphaFromSubstrateKTheory_r123.lean` in the parent repository, culminating in the kernel-verified capstone `substrate_level_projection_trace`. What is formalized is this finite-level projection-trace computation. The full classical identity $K_0(M_{3^\infty}) \cong \mathbb{Z}[1/3]$ (Glimm [1]) is *not* formalized: K_0 is not constructed in Lean. We cite it as established classical mathematics and use only the formalized fact that the realized finite-level trace values — which generate the classical range — lie in $\mathbb{Z}[1/3]$; this is all the obstruction argument below consumes.

Theorem 4.2 (π is not a $\mathbb{Z}[1/3]$-ratio; Cohen, r332, 2026-09-07)**.**

```
theorem pi_not_ktheoretic_ratio
    {a b : ℝ} (ha : MemZ13 a) (hb : MemZ13 b) (h0 : a ≠ 0) :
    b / a ≠ Real.pi
```

where `MemZ13 x` $:= \exists (m : \mathbb{Z})(k : \mathbb{N}), x = m/3^k$.

Proof (as recorded in the source). Immediate: $\mathbb{Z}[1/3] \subseteq \mathbb{Q}$, so any ratio of $\mathbb{Z}[1/3]$-elements is rational; π is not. □

The Lean source is at `PF/AlphaL5PiScalingObstruction_r332.lean`, with the kernel-audit verdict {`propext`, `Classical.choice`, `Quot.sound`}. The r332 file additionally records `alpha_ns_div_alpha_rh_eq_pi` (an `rfl` after unfolding the two α-definitions), `pi_not_ktheoretic_ratio` non-vacuity, and a remark that the $a \neq 0$ hypothesis is redundant (with a stated proof).

Corollary 4.3 (Joint statement)**.** *Combining Theorem 3.5, Theorem 4.1, and Theorem 4.2:*

(1) The substrate T_∞ is uniquely determined — up to $*$-isomorphism — by the four fields of `Substrate3Inf`.

(2) Its complete classifying invariant range is $\mathbb{Z}[1/3] \subseteq \mathbb{Q}$.

(3) Consequently, no $\alpha \in \mathbb{R} \setminus \mathbb{Q}$ is in the range of the substrate's trace pairing, and no ratio $\alpha_i/\alpha_j = \pi$ (equivalently, $\alpha_{\mathrm{NS}}/\alpha_{\mathrm{RH}}$ in the Principia Fractalis notation) can be reconstructed from that pairing.

(4) Any derivation of the framework's α-skeleton must therefore proceed through a channel other than the K-theoretic trace pairing. The substrate alone does not supply such a derivation.

The four α-values that are *not* excluded by conjuncts (2) and (3) alone are the members of $\mathbb{Q}$ appearing in the program's enumeration: $\alpha_{\text{Poincaré}} = 1$, $\alpha_{\mathrm{NP}} = \varphi + 1/4$ (which lies in $\mathbb{Q}$ only via its rational offset — φ itself is irrational, so $\alpha_{\mathrm{NP}} \notin \mathbb{Z}[1/3]$ unless the ambiguity $\varphi + 1/4 = 1/4 + \varphi$ is broken artificially), $\alpha_{\mathrm{RH}} = 3/2$, and (with reservation as just noted) $\alpha_{\mathrm{P}} = \sqrt{2}$ likewise irrational. In fact, at most two of the nine values ($\alpha_{\text{Poincaré}}$ and α_{RH}) actually lie in $\mathbb{Z}[1/3]$; the remaining seven, including all π-multiples and $\sqrt{\ }$-radicals and φ-related values, are provably outside the substrate's invariant reach by Theorem 4.2.

The parent codebase's own audit apparatus [7] has verified in a separate report that all eight structural laws underpinning the α-skeleton uniqueness result (Cohen, r128) are *independent* in the technical sense: none is derivable from the formalized substrate assumptions through the tested channels. That report [7] concludes with the recommended standing sentence:

The nine α-values are uniquely determined by eight structural laws together with positivity and the external Perelman anchor. Those eight laws are independent assumptions: none is redundant, none is derivable from the framework's substrate, and the system is triangular — one constraint per value, with no over-determination. The α-skeleton is therefore a presentation *of the nine values, not a derivation of them.*

The present paper's joint statement (Corollary 4.3) is the substrate-side machine-checked companion to that sentence: the substrate is not a choice, but it also does not supply the α-values.

5 Proof architecture

The proof is organized into four proof-block arcs (with a fifth witness-construction arc, W), each corresponding to a self-contained sub-goal:

Arc W (5 cards): Construction of `substrate3Inf_TimelessFieldCompletion` — the witness that `TimelessFieldCompletion` satisfies `Substrate3Inf`. Uses Pablo Cohen's existing kernel-verified apparatus for the completion, the direct-limit trace, and Pablo's uniqueness theorem `substrate_UHF_trace_unique`.

Arc C1 (7 cards): The block-diagonal $*$-embedding $M_n(\mathbb{C}) \hookrightarrow M_{kn}(\mathbb{C})$ given by $x \mapsto x \otimes I_k$, packaged as a unital, injective, isometric $*$-algebra homomorphism. Includes the reindex through `Matrix.reindexAlgEquiv` to land in $M_{k\cdot n}(\mathbb{C})$ with the flattened index.

Arc C2 (12 cards): Noether–Skolem uniqueness for $M_n(\mathbb{C})$. Given two unital $*$-homomorphisms $\varphi, \psi : M_n(\mathbb{C}) \to M_{kn}(\mathbb{C})$, there exists a unitary $U \in M_{kn}(\mathbb{C})$ with $\varphi(x) = U\psi(x)U^*$ for all x. Proved via the matrix-unit calculus, an orthonormal-basis construction `phiONB`, and the change-of-basis unitary `unitaryOfONBpair`.

Arc C3 (5 cards): Universal property of the C^*-completion, star-upgraded. Given a coherent family of $*$-alg-homs from a dense tower, extend uniquely to a $*$-alg-hom on the ambient completion. Proved via `IsDenseInducing.extend` and `DenseRange.induction_on` for each of the seven `StarAlgHom` fields.

Arc C4 (11 cards + main): Level-preserving recursion (one-sided; no zig-zag, since both towers share the dimension sequence 3^k). Given two `Substrate3Inf` witnesses on A and B, construct a coherent family of level-k $*$-isos, apply Noether–Skolem inductively to force compatibility with tower inclusions, extend via C3 to $F : A \to B$ and $G : B \to A$, then verify $F \circ G = \mathrm{id}$ and $G \circ F = \mathrm{id}$ on the dense subalgebra to package as $A \simeq_{*\text{-alg}} B$. Specialization $B =$ `TimelessFieldCompletion` then gives Theorem 3.5.

5.1 Level-preserving intertwiner (C4.2)

The core inductive step of the level-preserving recursion is the following. Given a $*$-iso $\varphi_k : A_k \to B_k$ at some level k, we need to produce a $*$-iso $\varphi_{k+1} : A_{k+1} \to B_{k+1}$ such that the diagram

$$\begin{array}{ccc} A_k & \xrightarrow{\varphi_k} & B_k \\ {\scriptstyle \iota_A}\downarrow & & \downarrow{\scriptstyle \iota_B} \\ A_{k+1} & \xrightarrow{\varphi_{k+1}} & B_{k+1} \end{array}$$

commutes. A naive choice — pick any $*$-iso between the level-$k{+}1$ matrix algebras — fails to intertwine the inclusions. The correction is Noether–Skolem: the two paths $A_k \to B_{k+1}$

(via $\iota_B \circ \varphi_k$ and via any candidate $\varphi_{k+1} \circ \iota_A$), transported to $M_{3^k}(\mathbb{C}) \to M_{3^{k+1}}(\mathbb{C})$ via the tower matrix isos, are both unital $*$-homomorphisms; by Arc C2 they differ by conjugation by a unitary U; correcting the candidate φ_{k+1} by inner conjugation by U^* intertwines the diagram by construction.

The proof of the intertwining step uses only associativity of matrix multiplication, unitarity ($UU^* = U^*U = 1$), and the C2 identity. We take particular care in the Lean proof to avoid the `ring` and `ring_nf` tactics, which are unsound on non-commutative matrix multiplication; we use explicit `mul_assoc`, `one_mul`, `mul_one`, and `star_mul` chains in `calc` blocks.

6 Mathlib contributions

Six auxiliary results in the source are of independent interest and are candidates for upstream contribution to `mathlib4`:

1. `conjByUnitary` — given a unitary $U \in M_n(\mathbb{C})$, the map $x \mapsto UxU^*$ is a $*$-alg-equivalence $M_n(\mathbb{C}) \simeq_{*\text{-alg}} M_n(\mathbb{C})$; the inverse is $x \mapsto U^*xU$. Currently absent from mathlib; we provide a complete construction with all seven `StarAlgEquiv` field proofs, using only associativity and unitarity.

2. `reindexStarAlgEquiv` — upgrades `Matrix.reindexAlgEquiv` to preserve $\star$ (via `Matrix.conjTranspose_re` which is definitionally `rfl` when the same reindex equivalence is applied to both row and column axes).

3. `coe_iSup_of_directed_starSubalgebra` — mathlib provides the corresponding lemma for `NonUnitalStarSubalgebra` but not for the unital version; we fill the gap in ∼25 lines.

4. A C^*-algebraic direct-limit construction for the non-commutative ternary tower (`TimelessFieldRing` and its equipment) — mathlib's `Ring.DirectLimit` requires `CommRing` and does not adapt; the generic `DirectLimit` lacks metric structure.

5. A star lift for `UniformSpace.Completion` — mathlib has `UniformSpace.Completion.extensionHom` at the `RingHom` level but not at the $*$-algebra level. Our construction `TimelessFieldRingToCompletionStar` specializes the pattern and could be generalized to a mathlib helper `IsDenseInducing.extendStarAlgHom`.

6. The `IsTracialLinearFunctional` predicate itself, with an associated uniqueness theorem for the 3^∞ substrate — mathlib has zero coverage of tracial states.

Each of the first two are less than 50 lines and could be submitted as standalone mathlib PRs. Items 3–6 would require broader discussion with the mathlib community regarding scope and naming conventions.

7 Kernel-audit verdict

Building `PF.SubstrateRigidity` against the pinned `mathlib4` (`v4.24.0-rc1`) yields the following audit block, output by the seven `#print axioms` directives at the tail of the file:

```
T_infinity_rigidity : [propext, Classical.choice, Quot.sound]
Substrate3Inf.trace_unique : [propext, Classical.choice, Quot.sound]
substrate3Inf_iso : [propext, Classical.choice, Quot.sound]
substrate3Inf_TimelessFieldCompletion : [propext, Classical.choice, Quot.sound]
Substrate3Inf.connect_iso : [propext, Classical.choice, Quot.sound]
Substrate3Inf.connect_unital : [propext, Classical.choice, Quot.sound]
IsTracialLinearFunctional : [propext, Classical.choice, Quot.sound]
```

No `sorryAx`, no `ofReduceBool`, no project axioms. The three axioms shown are the standard Lean kernel axioms (propositional extensionality, classical choice, and quotient soundness), assumed by `mathlib4` throughout. The audit directives are live in the source at commit `398e3dfd` (lines 2504–2510). A prose comment in the file's header, carried over from an earlier development stage, still describes this block as "commented out while sorries remain"; that description is *stale* — the sorries were discharged and the block emits the verdict above. We note it so a reader scanning the header is not misled.

8 Place in Principia Fractalis

This result should not be read in isolation from the research program in which the substrate object T_∞ was constructed. *Principia Fractalis* [6] is Pablo Cohen's long-running (2024–present) research program, presenting a candidate substrate-level framework based on a *Fractal Resonance Function*

$$R_f(\alpha, s) \;=\; \sum_{n \geq 1} \frac{e^{i\pi\alpha\, D_3(n)}}{n^s}$$

(where D_3 is the base-3 digit sum), with the Timeless Field T_∞ — the object formalized here as `TimelessFieldCompletion` — serving as its algebraic substrate. The program posits that a nine-parameter α-skeleton, subject to eight structural laws, projects into a family of physical and mathematical sectors. The framework's Millennium-problem material is conditional reduction, not resolution; none of it is used in this paper [5].

The Principia Fractalis codebase, hosted at `github.com/FractalDevTeam/Principia-Fractalis`, comprises approximately 4.3×10^5 lines of Lean 4 with zero `sorry` declarations in the main build (excluding the `PF.SubstrateRigidity` module presented here, which was not yet wired into the top-level import when this paper was written). It includes over 10^4 machine-verified theorems, a book manuscript of 36 chapters and 13 appendices, and 15 accompanying research papers.

Where the joint statement lands in the program

The paired result (Corollary 4.3) discharges the "central theorem" specified in §1 of the program's own steering document `UNIFIED_THEORY_PROOF_PROGRAM.md` (dated 2026-09-09):

> *The Timeless Field substrate T_∞ is the UHF algebra of supernatural type 3^∞ [...]. Any C^*-algebra satisfying the substrate axioms (ternary directed system, norm-preserving connecting maps, C^*-identity) is $*$-isomorphic to it.*

Theorem 3.5 is that discharge, formalized. In the language of Stone–von Neumann or GNS it is a *rigidity theorem*: the substrate is not a modelling choice — it is the unique object of its class, up to $*$-isomorphism, characterized by four Lean fields.

The pairing with Theorem 4.2 sharpens the reading and, in our view, matches what the program has actually established about its substrate axis. What follows is the reading we recommend for downstream discussion:

The two-axis picture

Principia Fractalis has *two* independent axes at its foundation:

Axis 1 — the substrate. The C^*-algebraic object T_∞. Kernel-verified as of the present work to be uniquely determined by its ternary UHF structure and its unique tracial linear functional. This paper's contribution.

Axis 2 — the α-skeleton. The nine real-valued parameters $\alpha_{\text{Poincaré}} = 1$, $\alpha_{\text{Hodge}} = \varphi$, $\alpha_{\text{RH}} = 3/2$, $\alpha_{\text{YM}} = 2$, $\alpha_{\text{BSD}} = 3\pi/4$, $\alpha_{\text{NS}} = 3\pi/2$, $\alpha_{\text{P}} = \sqrt{2}$, $\alpha_{\text{NP}} = \varphi + 1/4$, $\alpha_{\text{QG}} = \sqrt{2\pi}$. Kernel-verified in the parent project (Cohen, r128) to be uniquely determined *given* eight structural laws and the Perelman anchor $\alpha_{\text{Poincaré}} = 1$. The eight laws are, per the rigidity audit [7], *independent* assumptions.

Theorem 4.2 establishes that the two axes are not linked through the substrate's K-theoretic channel: seven of nine α-values are irrational (and one further ratio, $\alpha_{\text{NS}}/\alpha_{\text{RH}}$, equals π), placing them outside $\mathbb{Z}[1/3]$; the substrate's classifying invariant has image exactly $\mathbb{Z}[1/3]$. Any bridge between the two axes must therefore be built at a higher layer than the substrate — through, for instance, sector-specific interface theorems (per the program's own directive, §5).

What Theorem 3.5 *does not* entail

Directly and without hedging:

- **No α-value is derived by the substrate.** Theorem 4.2 makes this obstruction explicit through the K-theoretic channel; the rigidity audit [7] makes it explicit through the direct-membership channel; between them, the "α-values as substrate consequences" reading is closed. The nine α-values remain, in the audit's own recommended phrasing, "a presentation of the nine values, not a derivation of them."
- **No Clay Millennium Problem is resolved.** The rigidity theorem does not touch the Riemann Hypothesis, Yang–Mills mass gap, Navier–Stokes regularity, BSD, Hodge, or P versus NP. The parent program's own machine-verified negative results (chapter-level refutations recorded in the codex tree; see [6]) are unaffected either way.
- **No sector correspondence is established.** The program's Section 5 obligation (per-sector interface theorems bridging the substrate to standard mathematical objects) is untouched by this result. That work is ongoing; the substrate layer is now stable enough to serve as an anchor for it, but the interfaces themselves must be built.

What the joint statement is worth

We claim, narrowly and, we believe, defensibly:

1. Principia Fractalis's substrate is a well-defined, unique-up-to-isomorphism C^*-algebraic object, kernel-certified.
2. The substrate's classifying invariant provably underdetermines the framework's α-skeleton, and any derivation of the α-values must be established at a higher layer.
3. Whatever downstream mathematical or physical content the program eventually establishes will rest on a substrate that is not a modelling degree of freedom, and on an α-skeleton whose grounding is a subject for continued research.

We regard this as the honest scientific summary of what has been kernel-verified about the substrate axis of Principia Fractalis through 2026-09-10. It is neither a headline claim about the framework's Clay-adjacent aspirations, nor a dismissal of them — it is the substrate-side theorem plus the substrate-side obstruction, offered as a joint statement whose downstream implications remain for the sector-interface work to determine.

9 Related work

Formalization of C*-algebras. Buzzard, Nash, Riccardo, and others have contributed substantially to `mathlib4`'s C^*-algebra infrastructure, culminating in the current `CStarAlgebra` class hierarchy [8]. Our work builds on this infrastructure and, where it falls short, fills the gaps in a manner consistent with mathlib conventions. To our knowledge, no prior Lean formalization has treated inductive limits of matrix algebras in a C^*-algebraic (as opposed to purely algebraic) manner.

Formalization of classification theorems. Alper, Franke, and others have formalized the classification of finite simple groups in various fragmentary states; Kepler's conjecture [10] and the four-color theorem [11] are the standard classification-adjacent success stories. Most recently, Anthropic's formalization of Fermat's Last Theorem [9] demonstrated the practical viability of large-scale language-model-assisted formalization, from which the present work adopts the statement-card discipline.

Bratteli diagrams and Elliott invariants. A full formalization of Elliott's classification for AF C^*-algebras [2] would be a much more ambitious undertaking than the present specialization; we regard the present work as an initial move in that direction, with the UHF 3^∞ specialization serving as a proof-of-concept.

10 Future work

The formalization is a specialization; the natural generalizations are:

1. UHF algebras of arbitrary supernatural number $\prod_p p^{n_p}$. Requires formalizing supernatural numbers themselves (a natural type constructor), Bratteli diagrams, and re-parameterizing the tower.

2. AF (C^*)-algebras — Elliott's full 1976 classification via the ordered K_0 group. Substantially more machinery; a plausible medium-term target.

3. Extending `IsTracialLinearFunctional` to include positivity and star-hermitianness — currently omitted (see Remark 3.3) but readily added. Would tighten the semantic content of the trace hypothesis to match the standard definition.

4. Building the sector-interface theorems in the parent Principia Fractalis program on top of the now-stable substrate layer (per the program's own directive, §5).

11 Reproducibility and availability

The two headline results are each a single-file formalization: the substrate rigidity theorem `T_infinity_rigidity` (with its derived `Substrate3Inf.trace_unique`) in `PF/SubstrateRigidity.lean` (approximately 2,500 lines), and the obstruction theorem `pi_not_ktheoretic_ratio` in `PF/AlphaL5PiScalingO` Both are present in the public repository `FractalDevTeam/Principia-Fractalis` at commit `398e3dfd` (2026-09-29), reachable on the public `master` branch (`origin/master`). To build the two headline theorems and reproduce their axiom audit:

```
git clone https://github.com/FractalDevTeam/Principia-Fractalis.git
cd Principia-Fractalis && git checkout 398e3dfd
cd PF_Lean4_Code
# Cap to two cores: Lake 5.0.0 exposes no -j flag, and the
# C*-algebra build is memory-bound, so more than one worker can
# exhaust RAM (OOM) on memory-constrained hardware.
```

```
taskset -c 0,1 lake build PF.SubstrateRigidity
taskset -c 0,1 lake build PF.AlphaL5PiScalingObstruction_r332
```

Each file's `#print axioms` directive emits exactly

```
'...T_infinity_rigidity' depends on axioms:  [propext, Classical.choice, Quot.sound]
'...Substrate3Inf.trace_unique' depends on axioms:  [propext, Classical.choice, Quot.sound]
'...pi_not_ktheoretic_ratio' depends on axioms:  [propext, Classical.choice, Quot.sound]
```

with no `sorry`, no `native_decide`, and no project axioms. The pinned toolchain is `leanprover/lean4:v4.24.0` with `mathlib4` at its `lakefile.lean`-pinned version.

Independent verification in a second proof assistant. The substrate's finite-dimensional core is independently verified in Rocq (Coq) 9.1.0, in the `PF_Coq_Code/PF_Real` development (four files — `Base3Tower`, `MatrixTraceFaithful`, `MatrixTracialUnique`, `WeylAveraging` — comprising approximately 125 declarations over `mathcomp`, as present at the reproducibility commit `398e3dfd`). This covers the matrix-level content that arcs C1–C2 consume: the ternary tower embeddings $M_{3^k}(F) \hookrightarrow M_{3^{k+1}}(F)$ with their trace scaling $\mathrm{tr}(\mathrm{embed}\, A) = 3\,\mathrm{tr}(A)$, the partial trace, trace faithfulness, and uniqueness of the normalized tracial functional. Every `Print Assumptions` in these files reports `Closed under the global context` — no axioms. The completion-level theorem `T_infinity_rigidity` is formalized in Lean only; the Rocq development corroborates the finite-dimensional infrastructure on which its matrix arcs rest, giving the substrate's finite core an independent two-assistant check.

Authorship convention. The Lean source headers credit "Pablo Cohen + Claude Opus", recording the AI-assisted proof-engineering relationship described in the acknowledgments. Following the authorship norms of arXiv and of the mathlib community — under which a language model is a tool, not an author — this paper lists Pablo Cohen as sole author, with the language model credited in the acknowledgments. These are the same relationship under two conventions, not a discrepancy.

Acknowledgments

The author acknowledges the ORCID identifier `0009-0002-0734-5565` and thanks the mathlib community for the deep C^*-algebra infrastructure that made this work possible. The author thanks Anthropic for the compute infrastructure and the Claude Code / Claude Agents platform on which the coordinated proof-farming discipline was implemented, adopting patterns from Anthropic's own Fermat's Last Theorem formalization [9]. All mathematical decisions — theorem specifications, proof strategy, semantic scope — were the author's; the Lean 4 proof engineering was farmed via prover-agent sessions with human-authored statement cards and human-audited verdicts.

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A Full Lean signatures of the load-bearing declarations

For reference, the exact Lean 4 signatures of the seven kernel-audited declarations:

```
-- 1. The tracial linear functional predicate.
structure IsTracialLinearFunctional
    (A : Type*) [CStarAlgebra A] (φ : A →ℂ) : Prop where
  continuous : Continuous φ
  add : ∀x y : A, φ(x + y) = φx + φy
  smul : ∀(c : ℂ) (x : A), φ(c •x) = c * φx
  tracial : ∀x y : A, φ(x * y) = φ(y * x)
  unital : φ1 = 1

-- 2. The characterization structure (four fields; trace uniqueness is
-- a derived theorem, item 7, not a field -- r337, 2026-09-17).
structure Substrate3Inf (A : Type*) [CStarAlgebra A] where
  tower : ℕ→StarSubalgebra ℂA
  tower_matrix : ∀k, Nonempty
                 (tower k ≃⋆ₐ[ℂ] Matrix (Fin (3^k)) (Fin (3^k)) ℂ)
  tower_mono : ∀k, tower k ≤tower (k+1)
  tower_dense : Dense (((⨆ k, tower k : StarSubalgebra ℂA) : Set A))

-- 3. Auxiliary lemmas on the structure.
lemma Substrate3Inf.connect_unital
    {A : Type*} [CStarAlgebra A] (h : Substrate3Inf A) (k : ℕ) :
    (StarSubalgebra.inclusion (h.tower_mono k)) 1 = 1

lemma Substrate3Inf.connect_iso
    {A : Type*} [CStarAlgebra A] (h : Substrate3Inf A) (k : ℕ) :
    Isometry (StarSubalgebra.inclusion (h.tower_mono k))

-- 4. The general classification theorem.
theorem substrate3Inf_iso
    {A B : Type*} [CStarAlgebra A] [CStarAlgebra B]
    (hA : Substrate3Inf A) (hB : Substrate3Inf B) :
```

```
    Nonempty (A ≃⋆ₐ[ℂ] B)

-- 5. The witness on the concrete substrate.
noncomputable def substrate3Inf_TimelessFieldCompletion :
    Substrate3Inf TimelessFieldCompletion

-- 6. The main theorem.
theorem T_infinity_rigidity
    (A : Type*) [CStarAlgebra A]
    (h : Substrate3Inf A) :
    Nonempty (A ≃⋆ₐ[ℂ] TimelessFieldCompletion) :=
  substrate3Inf_iso h substrate3Inf_TimelessFieldCompletion

-- 7. Trace uniqueness, DERIVED (r337) -- not a hypothesis.
theorem Substrate3Inf.trace_unique
    {A : Type*} [CStarAlgebra A] (h : Substrate3Inf A) :
    ∃! τ: A →ℂ, IsTracialLinearFunctional A τ
```